

A coordinate-free formulation of the free energy asymptotics in singular learning theory

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Abstract

We rework the derivation of the asymptotic expansion of the Bayesian free energy in singular learning theory, following Watanabe [1], so that the statements and the limiting objects are free of local coordinates on the resolved parameter space. The key observation is that the leading coefficient of the asymptotic expansion of the partition function is a positive measure μ on the resolution manifold, definable directly as a Laurent coefficient of the zeta function of the model. Watanabe’s singular integral estimates (Theorems 4.6–4.9 of [1]) are then subsumed, at the level of statements, into a single kernel convergence theorem, valid for a general class of insertions, from which the limit law of the normalized evidence and the free energy formula

$$F_n = n\beta S_n + \lambda \log n - (m-1) \log \log n + F_n^R, \quad F_n^R \rightarrow -\log \mathcal{Y}(\xi) \text{ in law,}$$

follow by soft arguments. Local charts survive only inside proofs; no partition-of-unity consistency conditions, essential families of coordinates, or chart constants appear in any statement. Convergence in law is treated under the index $s = 2$ moment conditions of [1], and convergence of expectations under the index $s = 4$ conditions; for the latter, the only ingredient beyond the uniform integrability machinery of [1] is a matching lower bound on the evidence, which the kernel theorem supplies through a deterministic insertion.

1 Introduction

Let $q(x)$ be a true distribution, $p(x | w)$, $w \in W \subset \mathbb{R}^d$, a statistical model, $\varphi(w)$ a prior, and let $D_n = \{X_1, \dots, X_n\}$ be independent samples from q . The central object of singular learning theory is the *stochastic complexity* (minus log evidence) at inverse temperature $\beta > 0$,

$$F_n = -\log \int_W \prod_{i=1}^n p(X_i | w)^\beta \varphi(w) dw.$$

When the set of true parameters $W_0 = \{w : p(\cdot | w) = q\}$ is a point at which the Fisher information is nondegenerate, the Laplace approximation yields $F_n = n\beta S_n + \frac{d}{2} \log n + O_P(1)$. In singular models — mixtures, neural networks, reduced-rank regression, hidden Markov models — W_0 is a real analytic set with singularities, the Laplace approximation fails, and Watanabe’s theory [1] shows that instead

$$F_n = n\beta S_n + \lambda \log n - (m-1) \log \log n + O_P(1), \tag{1}$$

where $(-\lambda)$ is the largest pole of the zeta function $\zeta(z) = \int K(w)^z \varphi(w) dw$ of the model, m is its order, and S_n is the empirical entropy of q . When the prior is positive near W_0 , λ is the real log canonical threshold of the Kullback–Leibler function K and m its multiplicity.

The proof in [1] proceeds by resolution of singularities: a proper real analytic map $g : M \rightarrow W$ is chosen so that $K \circ g$ and the pulled-back prior are simultaneously normal crossing in each chart of a manifold M , and the partition function is then analyzed chart by chart. The chart-level analysis (Theorems 4.6–4.9 of [1]) is sharp but heavily coordinate-dependent: the limiting random variable is exhibited as a sum over an “essential family of local coordinates” of integrals involving chart constants γ_b , monomial weights y^μ , and a partition of unity, and the consistency of these data across charts must be arranged by hand (Theorem 6.5 of [1]).

The purpose of this note is to show that both the statements and the limit objects can be made coordinate-free, in the following precise sense.

- (i) The leading coefficient of the asymptotic expansion of the partition function is a finite *positive measure* μ on M , definable in one line as a Laurent-type limit of the zeta function (Definition 3.1); its support Σ is the “essential locus” of the resolution. All chart constants and consistency conditions are absorbed into μ .
- (ii) Watanabe’s singular integral estimates are replaced by a single *kernel theorem* (Theorem 5.2): the renormalized functionals

$$\Lambda_n[F] = \frac{(m-1)! n^\lambda}{(\log n)^{m-1}} \int_M F(nK_g(u), u) \nu(du)$$

converge, with a quantitative rate, to $\int_0^\infty \int_\Sigma F(t, u) t^{\lambda-1} d\mu(u) dt$, for arbitrary insertions $F(t, u)$ that are C^1 in u with exponential decay in t . Watanabe’s insertions K^p , and the insertions needed later in the theory (posterior moments, fluctuation functionals), are special cases; “general p ” costs nothing.

- (iii) The limit law of the evidence and the free energy formula (1) follow from (i)–(ii) by the continuous mapping theorem: the limiting random variable is

$$\mathcal{Y}(\xi) = \frac{1}{(m-1)!} \int_0^\infty \int_\Sigma t^{\lambda-1} e^{-\beta t + \beta \sqrt{t} \xi(u)} d\mu(u) dt,$$

a functional of the pair $(\mu, \xi|_\Sigma)$ alone, where ξ is the limiting Gaussian process of the standard form of the log likelihood ratio. The pair depends on the chosen resolution; the law of $\mathcal{Y}(\xi)$ does not (Remark 8.1).

Charts are not eliminated: they persist inside the proofs, where the exact evaluation of monomial state densities is performed, and for these local calculations we cite [1] rather than repeat them. What disappears is every chart-dependent statement, together with the bookkeeping needed to glue chart-level results.

Sections 3–6 treat convergence in law, which requires only the index $s = 2$ moment conditions of [1]; Section 7 derives the corresponding statements in expectation under the index $s = 4$ conditions. There the upper-bound-only character of the toolkit must be supplemented by a matching lower bound on the evidence; the point of the coordinate-free framework is that this bound is not a new chart computation (the role played by [1, Thms. 4.8, 4.10]) but an

application of the kernel theorem to a deterministic insertion, combined with the positivity $\mu(M) > 0$.

The point of view that the coefficients of the asymptotic expansion of a Laplace-type integral are distributions supported on the critical locus, obtained as Laurent coefficients of $\int f^z \varphi$, with orders controlled by the drop in pole order, is classical in singularity theory; see Arnold–Gusein-Zade–Varchenko [3], Malgrange [5], and, for the complex-analytic counterpart, Barlet [6]. Section 8 contains precise pointers.

2 Background and standing assumptions

This section collects, with references to [1], the inputs we take as given: the analytic hypotheses on the model, the resolution package, the standard form of the log likelihood ratio, and the zeta function. A reader familiar with the main results of singular learning theory but not with the details of [1] should find here everything needed for the later sections.

2.1 The model and the fundamental conditions

Let $q(x)$ and $p(x \mid w)$, $w \in W \subset \mathbb{R}^d$, be probability densities on $\mathbb{R}^N$ with common support, W compact, and let

$$K(w) = \int q(x) \log \frac{q(x)}{p(x \mid w)} dx, \quad W_0 = \{w \in W : K(w) = 0\} \neq \emptyset,$$

be the Kullback–Leibler function and the set of true parameters. Write $f(x, w) = \log(q(x)/p(x \mid w))$, so that $K(w) = \mathbb{E}_X[f(X, w)]$. Given n independent samples from q , the *log likelihood ratio function* is $K_n(w) = \frac{1}{n} \sum_{i=1}^n f(X_i, w)$, and the *normalized evidence* and *normalized free energy* at inverse temperature $\beta > 0$ are

$$Z_n^0 = \int_W e^{-n\beta K_n(w)} \varphi(w) dw, \quad F_n^0 = -\log Z_n^0, \quad (2)$$

related to F_n by $F_n = n\beta S_n + F_n^0$, where $S_n = -\frac{1}{n} \sum_i \log q(X_i)$.

Assumption 2.1 (Fundamental conditions; [1, Def. 6.1, Def. 6.3]). (I) (index s , an even integer ≥ 2) The map $w \mapsto f(\cdot, w)$ extends to an $L^s(q)$ -valued complex analytic function on an open set $W^{(\mathbb{C})} \subset \mathbb{C}^d$ containing W ; the envelope $M(x) = \sup_{w \in W^{(\mathbb{C})}} |f(x, w)|$ lies in $L^s(q)$; and $\int M(x)^2 \sup_{K(w) \leq \epsilon} p(x \mid w) dx < \infty$ for some $\epsilon > 0$.

(II) $W = \{w : \pi_1(w) \geq 0, \dots, \pi_k(w) \geq 0\}$ with π_j real analytic, and $\varphi = \varphi_1 \varphi_2$ with $\varphi_1 > 0$ smooth and $\varphi_2 \geq 0$ real analytic.

Condition (I) at a larger even index implies it at any smaller one ([1, Rem. 6.1(7)]). Sections 3–6 use only $s = 2$, which is assumed by default throughout; Section 7 strengthens this to $s = 4$, stated there explicitly. Condition (I) is the analytic input driving the empirical process theory; condition (II) rules out pathological (flat but nonanalytic) priors and boundaries. We refer to [1, §6.1] for discussion.

2.2 The resolution package

The following consequence of Hironaka's theorem [2] is the geometric backbone; we fix its data once and for all.

Theorem 2.2 (Resolution package; [1, Thm. 6.5]). *Under Assumption 2.1 there exist $\epsilon \in (0, 1)$, a d -dimensional real analytic manifold containing a compact subset M , and a proper real analytic map g from that manifold onto an open neighborhood of $W_\epsilon := \{w \in W : K(w) \leq \epsilon\}$, with $M = g^{-1}(W_\epsilon)$, such that M is covered by finitely many charts $U_\alpha \cong [0, b]^d$ (nonnegative orthants) in which:*

- (1) $K_g := K \circ g$ is a normal crossing monomial, $K_g(u) = u^{2k} = u_1^{2k_1} \cdots u_d^{2k_d}$, with nonnegative integers $k_j = k_j^{(\alpha)}$;
- (2) the pullback of the prior, $\nu(du) := \varphi(g(u)) |g'(u)| du$, has the form $\nu(du) = \phi(u) u^h du$ with nonnegative integers $h_j = h_j^{(\alpha)}$ and $\phi \in C^\infty$, $\phi \geq c > 0$ on $[0, b]^d$;
- (3) there is a smooth partition of unity $\{\sigma_\alpha\}$ with $\sum_\alpha \sigma_\alpha = 1$ on M , $\sigma_\alpha > 0$ on $[0, b)^d$ and $\text{supp } \sigma_\alpha = [0, b]^d$; and for every integrable H ,

$$\int_{W_\epsilon} H(w) \varphi(w) dw = \int_M H(g(u)) \nu(du) = \sum_\alpha \int_{[0, b]^d} H(g(u)) \sigma_\alpha(u) \phi(u) u^h du. \quad (3)$$

We emphasize that the measure ν on M and the function K_g are globally defined; only their normal crossing presentations are chart data.

2.3 The standard form and the process ξ_n

Theorem 2.3 (Standard form; [1, Thm. 6.1, Main Thm. 6.1]). *In each chart there is an $L^2(q)$ -valued real analytic function $a(x, u)$ with*

$$f(x, g(u)) = a(x, u) u^k, \quad \mathbb{E}_X[a(X, u)] = u^k.$$

Consequently, with

$$\xi_n(u) = \frac{1}{\sqrt{n}} \sum_{i=1}^n (\mathbb{E}_X[a(X, u)] - a(X_i, u)), \quad (4)$$

the log likelihood ratio takes the standard form

$$K_n(g(u)) = K_g(u) - \frac{1}{\sqrt{n}} \sqrt{K_g(u)} \xi_n(u). \quad (5)$$

Three remarks. First, on signs: our ξ_n is the negative of [1, eq. (6.11)]; with the definition there, the standard form [1, eq. (6.10)] holds with a *plus* sign, so the two displays of [1] are mutually inconsistent as printed. The discrepancy is immaterial for results in law, since ξ and $-\xi$ are equal in law for a centred Gaussian process, but a definite convention matters once $a(x, u)$ enters expressions linearly (Remark 8.3); ours agrees with that of ψ_n in (6) below. Second, on globality: since the charts are nonnegative orthants, $u^k \geq 0$, so u^k is the *nonnegative* square root $\sqrt{K_g(u)}$ in every chart; on overlaps the presentations therefore agree, and $a = f(\cdot, g(\cdot))/\sqrt{K_g}$ off the exceptional divisor, extended analytically across it, is a single global

function on $\mathbb{R}^N \times M$. Hence ξ_n is a globally defined random element of $C(M)$, and (5) holds on all of M . Third, on the outer region: on $\{w : K(w) > \epsilon\}$ one uses instead the classical normalization

$$\psi_n(w) = \sqrt{n} \frac{K(w) - K_n(w)}{\sqrt{K(w)}}, \quad (6)$$

which is well defined there.

Theorem 2.4 ([1, Thms. 5.8, 5.9, 6.2, 6.3]). *Under Assumption 2.1, ξ_n converges in law in $C(M)$ (with the supremum norm) to a Gaussian process ξ with mean zero and covariance $\mathbb{E}[\xi(u)\xi(v)] = \mathbb{E}_X[a(X, u)a(X, v)] - u^k v^k$; moreover $\mathbb{E}[\xi(u)^2] = 2$ whenever $K_g(u) = 0$. Similarly ψ_n converges in law in $C(\{K \geq \epsilon\})$, and $\sup_n \mathbb{E}[\sup_M |\xi_n|^2] < \infty$, $\sup_n \mathbb{E}[\sup_{K \geq \epsilon} |\psi_n|^2] < \infty$.*

Two citation notes: in [1], the identity $\mathbb{E}[\xi(u)^2] = 2$ is [1, Thm. 6.3], stated there under index $s = 4$; its proof, however, uses only $M \in L^2(q)$ and condition (I) of Assumption 2.1, and is therefore available at $s = 2$. The uniform second-moment bounds are [1, Thm. 5.8] applied with $s = 2$.

2.4 The zeta function and the invariants (λ, m)

For $\Phi \in C(M)$ define

$$\zeta_\Phi(z) = \int_M K_g(u)^z \Phi(u) \nu(du). \quad (7)$$

Chartwise this is $\sum_\alpha \int u^{2kz+h} \Phi \sigma_\alpha \phi du$; the integral converges absolutely and is holomorphic for $\operatorname{Re} z > -\lambda$, where

$$\lambda = \min_\alpha \min_{j: k_j \geq 1} \frac{h_j + 1}{2k_j}, \quad m = \max_{\alpha: \min_j = \lambda} \#\{j : \frac{h_j + 1}{2k_j} = \lambda\}. \quad (8)$$

(Coordinates with $k_j = 0$ do not constrain convergence and are assigned exponent $+\infty$.) By [1, Thm. 6.6], ζ_1 continues to a meromorphic function on $\mathbb{C}$ whose poles are negative rational and lie among the chartwise candidates in (8); that the largest pole is exactly $-\lambda$ and of order exactly m — i.e. that no cancellation occurs when the charts are summed — will follow from the positivity of the leading measure (Proposition 3.2(i)). Since $\int_W K^z \varphi dw - \zeta_1(z)$ is an integral over the compact set $\{K \geq \epsilon\}$, hence entire, (λ, m) agree with the invariants of the zeta function of the model on all of W ; in particular they do not depend on the resolution.

Definition 2.5. A chart α is *essential* if the minimum in (8) equals λ there and is attained by exactly m indices. In an essential chart we relabel coordinates as $u = (x, y)$, $x = (u_1, \dots, u_m)$ the *essential* directions (so $(h_j + 1)/(2k_j) = \lambda$, $k_j \geq 1$, for $j \leq m$) and $y = (u_{m+1}, \dots, u_d)$ the rest, for which the inequality is strict:

$$\mu'_j := h'_j - 2\lambda k'_j > -1 \quad (m+1 \leq j \leq d), \quad (9)$$

writing (k', h') for the y -part of the multi-indices. Note that no chart contains more than m indices attaining λ , by maximality of m in (8).

3 The leading measure μ

Definition 3.1. For $\Phi \in C(M)$ set

$$\mu[\Phi] := \lim_{\substack{\sigma \rightarrow -\lambda^+ \\ \sigma \in \mathbb{R}}} (\sigma + \lambda)^m \zeta_\Phi(\sigma).$$

The definition involves no charts. Its existence, and everything we shall need, is:

Proposition 3.2. (i) *The limit exists for every $\Phi \in C(M)$ and defines a finite positive Radon measure μ on M with $0 < \mu(M) < \infty$. In particular $(\sigma + \lambda)^m \zeta_1(\sigma) \rightarrow \mu(M) > 0$, so the pole of ζ_1 at $-\lambda$ has order exactly m : positivity forbids cancellation across charts.*

(ii) *In the notation of Definition 2.5, in each essential chart*

$$d\mu = \frac{\sigma_\alpha(0, y) \phi(0, y) y^{\mu'} dy}{2^m k_1 k_2 \cdots k_m} \quad \text{on the face } \{x = 0\}, \quad (10)$$

summed over essential charts; non-essential charts contribute zero.

(iii) $\Sigma := \text{supp } \mu$ *is the union, over essential charts, of the closed faces $\{x = 0\}$. It is compact, of codimension m , contained in $g^{-1}(W_0)$, and admits the chart-free characterization: $u_0 \in \Sigma$ if and only if for every neighborhood $U \ni u_0$ there is $\Phi \geq 0$ in $C(M)$ supported in U with $\mu[\Phi] > 0$; equivalently, if and only if $(\sigma + \lambda)^m \zeta_\Phi(\sigma)$ has a strictly positive limit as $\sigma \rightarrow -\lambda^+$ for every $\Phi \geq 0$ continuous with $\Phi(u_0) > 0$ — a pole of order exactly m when Φ is smooth enough for ζ_Φ to continue meromorphically.*

Proof. Write $\varepsilon = \sigma + \lambda > 0$. Fix a chart and put $\Psi = \Phi \sigma_\alpha \phi \in C([0, b]^d)$.

One-variable identity. For $\psi \in C[0, b]$ and $\kappa \geq 1$, substituting $v = (x/b)^{2\kappa\varepsilon}$,

$$2\kappa\varepsilon \int_0^b x^{2\kappa\varepsilon-1} \psi(x) dx = b^{2\kappa\varepsilon} \int_0^1 \psi(b v^{1/(2\kappa\varepsilon)}) dv \xrightarrow{\varepsilon \rightarrow 0^+} \psi(0) \quad (11)$$

by dominated convergence, since $v^{1/(2\kappa\varepsilon)} \rightarrow 0$ for each $v \in (0, 1)$.

Essential charts. There $u^{2k\sigma+h} = (\prod_{j \leq m} x_j^{2k_j\varepsilon-1}) (\prod_{j > m} y_j^{\mu'_j+2k'_j\varepsilon})$, using $h_j = 2k_j\lambda - 1$ for essential j and (9). Applying (11) in each essential variable (for fixed y) and dominated convergence in y — the y -integrand is dominated by $\|\Psi\|_\infty \max(y^{\mu'}, y^{\mu'+2k'})$, integrable by (9) — gives

$$\varepsilon^m \int_{[0, b]^d} u^{2k\sigma+h} \Psi(u) du \longrightarrow \frac{1}{2^m k_1 \cdots k_m} \int_{[0, b]^{d-m}} \Psi(0, y) y^{\mu'} dy.$$

Non-essential charts. If the chart minimum λ_α exceeds λ , the chart integral is bounded near $\sigma = -\lambda$ (it converges absolutely for $\text{Re } z > -\lambda_\alpha$), so $\varepsilon^m \cdot (\text{chart integral}) \rightarrow 0$. If $\lambda_\alpha = \lambda$ with multiplicity $m_\alpha < m$, the computation just performed (with m_α in place of m) shows $\varepsilon^{m_\alpha} \cdot (\text{chart integral})$ converges, hence again $\varepsilon^m \cdot (\text{chart integral}) \rightarrow 0$.

Summing the finitely many charts proves existence of the limit and formula (10). Positivity of $\Phi \mapsto \mu[\Phi]$ is immediate from the definition, since $\zeta_\Phi(\sigma) \geq 0$ for $\Phi \geq 0$ and real $\sigma > -\lambda$; applying positivity to $\|\Phi\|_\infty \pm \Phi$ gives $|\mu[\Phi]| \leq \|\Phi\|_\infty \mu[1]$, and the Riesz representation theorem (or formula (10) directly) yields the measure. Since $\phi \geq c > 0$ and $\sigma_\alpha > 0$ on $[0, b)^d$, the density in (10) is strictly positive on a dense open subset of each essential face; hence $\mu(M) > 0$, each

essential face lies in $\text{supp } \mu$, and $\text{supp } \mu$ is exactly the union of the closed essential faces. On such a face, $K_g = x^{2k} y^{2k'} = 0$ because every essential $k_j \geq 1$; thus $\Sigma \subseteq g^{-1}(W_0)$. The characterization in (iii) restates $u_0 \in \text{supp } \mu$: the computation above shows that the one-sided limit exists, is finite, and equals $\mu[\Phi]$. $\square$

Remark 3.3 (Dictionary with [1]). Watanabe's Theorem 4.6 computes, for tuned exponents $(h_j+1)/(2k_j) \equiv \lambda$ on $[0, b]^r$, the state density $v(t) = \int \delta(t - a x^{2k}) x^h dx = \gamma_b t^{\lambda-1} a^{-\lambda} (\log(ab^{2|k|}/t))^{r-1}$ with

$$\gamma_b = \frac{b^{|h|+r-2|k|\lambda}}{2^r (r-1)! k_1 \cdots k_r}.$$

With $r = m$ and the exponents tuned, $|h| + m - 2|k|\lambda = 0$, so $(m-1)! \gamma_b = (2^m k_1 \cdots k_m)^{-1}$, matching (10): chartwise, $d\mu = (m-1)! \gamma_b \sigma_\alpha(0, y) \phi(0, y) y^{\mu'} dy$. In the same spirit, an inverse Mellin transform ([1, Ex. 4.7]) converts Definition 3.1 into the leading term of the state density: for $\Phi \in C^\infty(M)$, the pushforward $(K_g)_*(\Phi \nu)$ has density

$$v_\Phi(t) = \mu[\Phi] \frac{t^{\lambda-1} (\log \frac{1}{t})^{m-1}}{(m-1)!} (1 + o(1)), \quad t \downarrow 0.$$

(For merely continuous Φ the asymptotic persists, but requires an equidistribution argument on the level sets rather than the exact Mellin pair; it is not used below.) Thus μ is precisely “the distribution formed by the leading coefficients of the expansion,” and the consistency conditions among charts and partitions of unity that [1, Thm. 6.5] must arrange play no role: μ is defined without them.

Remark 3.4. For Φ smooth, ζ_Φ is meromorphic near $-\lambda$ (same continuation as [1, Thm. 6.6]) and $\mu[\Phi]$ is its top Laurent coefficient at $-\lambda$. For merely continuous Φ the continuation need not exist, but Definition 3.1 does not require it: the real one-sided limit is computed directly in Proposition 3.2. Positivity holds only for this top coefficient at the smallest exponent; the subleading coefficients are genuinely signed distributions.

4 Two estimates

Fix $A \geq 0$ and $\beta > 0$ and set $w_A(t) := (1+t)^A e^{-\beta t/2}$ for $t \geq 0$. Call a measurable $G : [0, \infty) \rightarrow \mathbb{R}$ *admissible* if $|G| \leq w_A$. Constants C below depend only on (A, β) and the fixed resolution data, never on n , G , or the test functions. We write

$$r_n := \begin{cases} n^{-\lambda} (\log n)^{m-2}, & m \geq 2, \\ n^{-\lambda-\delta} (\log n)^{M_0}, & m = 1, \end{cases}$$

for constants $\delta > 0$ and $M_0 \leq d-1$ determined by the resolution and specified in the proof of Lemma 4.2.

Lemma 4.1 (A priori bound). *For every admissible G and $\Phi \in C(M)$, and $n > e$,*

$$\left| \int_M G(nK_g(u)) \Phi(u) \nu(du) \right| \leq C \|\Phi\|_\infty \frac{(\log n)^{m-1}}{n^\lambda}.$$

More precisely, a single chart with data (k, h) , exponent λ_α and multiplicity m_α contributes at most $C \|\Phi\|_\infty (\log n)^{m_\alpha-1} n^{-\lambda_\alpha}$.

Proof. Chartwise. Group the chart coordinates into those attaining λ_α and the rest; this is the configuration of [1, Thm. 4.7], and the upper-bound half of its proof — push forward to the $t = nu^{2k}$ variable by [1, Thm. 4.6] and estimate — applies verbatim with the specific function $t^p e^{-\beta t}$ there replaced by any admissible G , since only the finiteness of $\int_0^\infty w_A(t) t^{\lambda_\alpha-1} (1 + |\log t|)^d dt$ is used. (Coordinates with $k_j = 0$ have exponent $+\infty$ and simply join the second group; note that every chart contains an index with $k_j \geq 1$, since $K_g \leq \epsilon < 1$ on M .) $\square$

Lemma 4.2 (Vanishing estimate). *Let $\Psi : [0, \infty) \times M \rightarrow \mathbb{R}$ be jointly measurable with $\Psi(t, \cdot) \in C^1(M)$ for each t ,*

$$N_A(\Psi) := \sup_{t,u} \frac{|\Psi(t,u)| + |\nabla_u \Psi(t,u)|}{w_A(t)} < \infty,$$

and suppose $\Psi(t, u) = 0$ for all $t \geq 0$ and all $u \in \Sigma$. Then for $n > e$,

$$\left| \int_M \Psi(nK_g(u), u) \nu(du) \right| \leq C N_A(\Psi) r_n.$$

(Here ∇_u and the C^1 norm are taken with respect to the fixed finite atlas; equivalently, any fixed Riemannian metric.)

Proof. Decompose by the partition of unity and treat charts separately.

Essential charts. The face $\{x = 0\}$ lies in Σ by Proposition 3.2(iii), so $\Psi(t, (0, y)) = 0$. The chart $[0, b]^d$ is convex, so the segment from (x, y) to $(0, y)$ stays in the chart, and the mean value theorem gives

$$|\Psi(t, (x, y))| = |\Psi(t, (x, y)) - \Psi(t, (0, y))| \leq N_A(\Psi) w_A(t) \sum_{j \leq m} x_j.$$

Hence the chart integral is bounded by $C N_A(\Psi) \sum_{j \leq m} \int_{[0,b]^d} w_A(nu^{2k}) u^{h+e_j} du$. The shifted data $(k, h + e_j)$ has j -th exponent $\lambda + \frac{1}{2k_j}$ and all other exponents unchanged. If $m \geq 2$, the minimum is still λ , now of multiplicity $m - 1$; Lemma 4.1 (chart version) gives the bound $C N_A(\Psi) (\log n)^{m-2} n^{-\lambda}$. If $m = 1$, the new minimum is at least $\lambda + \delta_\alpha$ with $\delta_\alpha := \min(\frac{1}{2k_j}, \min_{i > m} \frac{h'_i + 1}{2k'_i} - \lambda) > 0$, and the chart bound is $C N_A(\Psi) (\log n)^{d-1} n^{-\lambda-\delta_\alpha}$.

Non-essential charts. No vanishing is needed: Lemma 4.1 gives $(\log n)^{m_\alpha-1} n^{-\lambda_\alpha}$ with either $\lambda_\alpha = \lambda$, $m_\alpha \leq m - 1$ (absorbed into $(\log n)^{m-2} n^{-\lambda}$) or $\lambda_\alpha > \lambda$ (absorbed into either rate, since $n^{-(\lambda_\alpha-\lambda)}$ beats any power of $\log n$; note $m = 1$ forces $\lambda_\alpha > \lambda$ in every non-essential chart).

Take $\delta := \min_\alpha \delta_\alpha \wedge \min\{\lambda_\alpha - \lambda : \lambda_\alpha > \lambda\}$ and $M_0 := d - 1$. $\square$

Remark 4.3. Lemma 4.2 is the precise sense in which “the coefficient distributions one order below the top have order at most one”: subtracting the value on Σ costs one derivative of the test function and gains one power of $\log n$ (or, when $m = 1$, a power $n^{-\delta}$). For $\Psi(t, u) = G(t)\Phi(u)$ with $\Phi \in C^\infty$ vanishing on Σ , the same chart computation shows that ζ_Φ has a pole of order at most $m - 1$ at $-\lambda$.

5 The kernel theorem

Definition 5.1. For jointly measurable $F : [0, \infty) \times M \rightarrow \mathbb{R}$ with $F(t, \cdot) \in C^1(M)$ for each t and $N_A(F) < \infty$, define

$$\Lambda_n[F] := \frac{(m-1)! n^\lambda}{(\log n)^{m-1}} \int_M F(nK_g(u), u) \nu(du), \quad \Lambda[F] := \int_0^\infty \int_\Sigma F(t, u) t^{\lambda-1} d\mu(u) dt.$$

Theorem 5.2 (Kernel theorem). *There exist C and, when $m = 1$, constants $\delta' > 0$, $M'_0 \geq 0$, depending only on (A, β) and the resolution, such that for all such F and all $n > e$,*

$$|\Lambda_n[F] - \Lambda[F]| \leq C N_A(F) \times \begin{cases} (\log n)^{-1}, & m \geq 2, \\ n^{-\delta'} (\log n)^{M'_0}, & m = 1. \end{cases}$$

Proof. Write $\Lambda_n[F] = \frac{(m-1)! n^\lambda}{(\log n)^{m-1}} \sum_\alpha I_n^\alpha$ with $I_n^\alpha = \int_{[0, b]^d} F(nu^{2k}, u) \sigma_\alpha \phi u^h du$ (we suppress the chart identification in the second argument of F).

Non-essential charts. By Lemma 4.1, $|I_n^\alpha| \leq C N_A(F) (\log n)^{m_\alpha-1} n^{-\lambda_\alpha}$, which after normalization is $O(N_A(F) (\log n)^{-1})$, resp. $O(N_A(F) n^{-\delta'} (\log n)^{M'_0})$ when $m = 1$, as in the proof of Lemma 4.2.

Essential charts: restriction to the face. Set

$$J_n^\alpha := \int_{[0, b]^d} F(nx^{2k}y^{2k'}, (0, y)) \sigma_\alpha(0, y) \phi(0, y) x^h y^{h'} dx dy.$$

The integrand of $I_n^\alpha - J_n^\alpha$ is $\Psi(nu^{2k}, u) u^h$ where $\Psi(t, (x, y)) := F(t, (x, y)) \sigma_\alpha(x, y) \phi(x, y) - F(t, (0, y)) \sigma_\alpha(0, y) \phi(0, y)$ vanishes at $x = 0$ and satisfies $|\nabla_x \Psi| \leq C N_A(F) w_A(t)$ (product rule, using $\|\sigma_\alpha \phi\|_{C^1} \leq C$). Exactly as in the proof of Lemma 4.2 (mean value theorem in the convex chart, then Lemma 4.1 with data $(k, h + e_j)$),

$$|I_n^\alpha - J_n^\alpha| \leq C N_A(F) r_n.$$

Essential charts: exact evaluation. Fix y with $y^{2k'} > 0$ (the complement is ν -null on the chart). By [1, Thm. 4.6], applied with the m tuned exponents $(h_j + 1)/(2k_j) = \lambda$ and scale $a = n y^{2k'}$, the pushforward of $x^h dx$ on $[0, b]^m$ under $x \mapsto t = n y^{2k'} x^{2k}$ is

$$\gamma_b t^{\lambda-1} (n y^{2k'})^{-\lambda} \left(\log \frac{n y^{2k'} b^{2|k|}}{t} \right)^{m-1} dt \quad \text{on } (0, T_y), \quad T_y := n y^{2k'} b^{2|k|},$$

with γ_b as in Remark 3.3. Hence, by Fubini, and writing $c(t, y) := \log(y^{2k'} b^{2|k|}/t)$,

$$J_n^\alpha = \gamma_b n^{-\lambda} \int dy \sigma_\alpha(0, y) \phi(0, y) y^{\mu'} \int_0^{T_y} F(t, (0, y)) t^{\lambda-1} (\log n + c(t, y))^{m-1} dt,$$

where $y^{h'} (y^{2k'})^{-\lambda} = y^{\mu'}$. Expand the binomial $(\log n + c)^{m-1} = \sum_{j=0}^{m-1} \binom{m-1}{j} (\log n)^{m-1-j} c^j$.

The $j = 0$ term equals, after multiplying by the normalization $(m-1)! n^\lambda (\log n)^{1-m}$ and recalling $(m-1)! \gamma_b \sigma_\alpha(0, y) \phi(0, y) y^{\mu'} dy = d\mu|_\alpha$ from (10),

$$\int_0^\infty \int F(t, u) t^{\lambda-1} d\mu|_\alpha(u) dt - \underbrace{\int \int_{t \geq T_y} F t^{\lambda-1} d\mu|_\alpha dt}_{\text{truncation}}.$$

For the truncation, note $T_y \leq t$ iff $y^{2k'} \leq s := t/(nb^{2|k|})$. Choose δ_1 with $0 < \delta_1 < \min\{(\mu'_j + 1)/(2k'_j) : k'_j \geq 1\}$ (any $\delta_1 > 0$ if no $k'_j \geq 1$). The pointwise Chebyshev bound $\mathbf{1}\{y^{2k'} \leq s\} \leq (s/y^{2k'})^{\delta_1}$, valid off the null set $\{y^{2k'} = 0\}$, gives

$$\int_{\{y^{2k'} \leq s\}} y^{\mu'} dy \leq s^{\delta_1} \int_{[0,b]^{d-m}} y^{\mu' - 2\delta_1 k'} dy = C s^{\delta_1} \quad (s > 0),$$

the last integral being finite by the choice of δ_1 and (9). Hence the truncation is at most $C N_A(F) n^{-\delta_1} \int_0^\infty w_A(t) t^{\lambda + \delta_1 - 1} dt \leq C N_A(F) n^{-\delta_1}$. (At the corresponding point the proof of [1, Thm. 4.9] cites [1, Cor. 4.3], which is stated only for tuned exponents; the display above supplies the untuned case, without the logarithmic factor.) The terms $1 \leq j \leq m - 1$ are bounded by

$$\binom{m-1}{j} (\log n)^{m-1-j} \gamma_b n^{-\lambda} N_A(F) \iint w_A(t) t^{\lambda-1} |c(t, y)|^j y^{\mu'} dt dy,$$

and the double integral is finite because $\lambda > 0$, $\mu'_j > -1$, and w_A decays exponentially; after normalization these are $O(N_A(F)(\log n)^{-1})$.

Summing over the essential charts, the leading terms assemble to $\Lambda[F]$ by Proposition 3.2(ii), and all error terms are within the stated rates. $\square$

Corollary 5.3 (All insertions of Watanabe type). *For $F(t, u) = t^p e^{-\beta t} \Phi(u)$ with $p \geq 0$ and $\Phi \in C^1(M)$,*

$$\Lambda_n[F] \longrightarrow \frac{\Gamma(\lambda + p)}{\beta^{\lambda+p}} \mu[\Phi].$$

Remark 5.4 (On the exponent conventions). The exponent of n in Λ_n is λ for *every* p : Corollary 5.3 inserts the renormalized quantity $t^p = (nK_g)^p$. Watanabe's Theorem 4.7 inserts $K_g^p = (t/n)^p$ instead, producing the rate $n^{-(\lambda+p)}$; the two statements differ by the explicit factor n^{-p} and are equivalent. The renormalized convention is the one consumed downstream (posterior moments of nK_n , fluctuation functionals), which is why Theorem 5.2 is stated for general $F(t, u)$: nothing further will be needed from singular integral theory.

6 The limit law of the evidence and the free energy formula

Lemma 6.1 (Gradient moments). $\sup_n \mathbb{E} \left[\|\xi_n\|_{C^1(M)}^2 \right] < \infty$, where $\|\eta\|_{C^1(M)} := \sup_M |\eta| + \max_\alpha \max_j \sup_{U_\alpha} |\partial_{u_j} \eta|$.

Proof. The bound for $\sup_M |\xi_n|$ is Theorem 2.4. For the gradient: on (a complex neighborhood of) each closed chart, $u \mapsto a(\cdot, u)$ is $L^2(q)$ -valued analytic (Theorem 2.3), with Taylor coefficients $a_\gamma \in L^2(q)$ obeying the Cauchy estimates $\|a_\gamma\|_2 \leq \|M_a\|_2 / R^\gamma$ of [1, Rem. 5.5, eq. (5.11)]. Term-by-term differentiation then exhibits $u \mapsto \partial_{u_j} a(\cdot, u)$ as an $L^2(q)$ -valued analytic family on a slightly smaller polydisc, with summable coefficient norms. Since $\partial_{u_j} \xi_n$ is, up to sign, the centred empirical process of this family — differentiation under $\mathbb{E}_X$ being justified by the same estimates — [1, Thm. 5.8] with $s = 2$ gives $\mathbb{E}[\sup_{U_\alpha} |\partial_{u_j} \xi_n|^2] \leq C_{\alpha,j}$, uniformly in n . Sum over the finite atlas. $\square$

Definition 6.2. For $\eta \in C(M)$ set

$$\mathcal{Y}(\eta) := \frac{1}{(m-1)!} \int_0^\infty \int_\Sigma t^{\lambda-1} e^{-\beta t + \beta \sqrt{t} \eta(u)} d\mu(u) dt.$$

Lemma 6.3. $0 < \mathcal{Y}(\eta) < \infty$ for every $\eta \in C(M)$, and $\mathcal{Y}$ is locally Lipschitz for the supremum norm: for $\|\eta\|_\infty, \|\eta'\|_\infty \leq R$,

$$|\mathcal{Y}(\eta) - \mathcal{Y}(\eta')| \leq \frac{\beta \mu(M) e^{\beta R^2/2}}{(m-1)!} \left(\int_0^\infty t^{\lambda-\frac{1}{2}} e^{-\beta t/2} dt \right) \|\eta - \eta'\|_\infty.$$

Proof. By $\beta \sqrt{t} \eta \leq \beta t/2 + \beta \eta^2/2$, the integrand is at most $t^{\lambda-1} e^{-\beta t/2} e^{\beta R^2/2}$, giving finiteness; positivity holds because the integrand is strictly positive and $\mu(M) > 0$ (Proposition 3.2). The Lipschitz bound follows from $|e^a - e^{a'}| \leq e^{\max(a, a')} |a - a'|$ with $a = \beta \sqrt{t} \eta(u)$, absorbing $e^{\beta \sqrt{t} R} \leq e^{\beta t/2 + \beta R^2/2}$. $\square$

Theorem 6.4 (Limit law of the normalized evidence). *Under Assumption 2.1,*

$$\frac{n^\lambda}{(\log n)^{m-1}} Z_n^0 \longrightarrow \mathcal{Y}(\xi) \quad \text{in law.}$$

Proof. Split $Z_n^0 = Z_n^{(1)} + Z_n^{(2)}$ over W_ϵ and $\{K > \epsilon\}$.

Outer region. On $\{K > \epsilon\}$, by (6) and $2\sqrt{nK} \psi_n \leq nK + \psi_n^2$,

$$n\beta K_n = n\beta K - \beta\sqrt{nK} \psi_n \geq \frac{\beta}{2} (n\epsilon - \sup_{K>\epsilon} \psi_n^2),$$

so $\frac{n^\lambda}{(\log n)^{m-1}} Z_n^{(2)} \leq \frac{n^\lambda e^{-n\beta\epsilon/2}}{(\log n)^{m-1}} \exp(\frac{\beta}{2} \sup_{K>\epsilon} \psi_n^2)$, which tends to 0 in probability since $\sup_{K>\epsilon} |\psi_n|$ converges in law (Theorem 2.4); this is display (6.18) of [1] and is already coordinate-free.

Inner region. By (3) and the standard form (5), using $\sqrt{n} \sqrt{K_g(u)} = \sqrt{nK_g(u)}$,

$$Z_n^{(1)} = \int_M \exp(-n\beta K_g(u) + \beta \sqrt{nK_g(u)} \xi_n(u)) \nu(du) = \int_M F_n(nK_g(u), u) \nu(du),$$

where $F_n(t, u) := e^{-\beta t + \beta \sqrt{t} \xi_n(u)}$. Thus $\frac{n^\lambda}{(\log n)^{m-1}} Z_n^{(1)} = \frac{1}{(m-1)!} \Lambda_n[F_n]$. The bounds $\beta \sqrt{t} \xi_n \leq \beta t/2 + \beta \xi_n^2/2$ and $|\nabla_u F_n| = \beta \sqrt{t} |\nabla_u \xi_n| F_n$ give, with $A = \frac{1}{2}$,

$$N_{1/2}(F_n) \leq C_\beta (1 + \beta \|\xi_n\|_{C^1}) e^{\beta \|\xi_n\|_\infty^2/2},$$

which is bounded in probability by Lemma 6.1 and Markov's inequality. Theorem 5.2 therefore yields

$$\left| \frac{n^\lambda Z_n^{(1)}}{(\log n)^{m-1}} - \mathcal{Y}(\xi_n) \right| = \frac{|\Lambda_n[F_n] - \Lambda[F_n]|}{(m-1)!} \leq C N_{1/2}(F_n) \cdot o(1) \longrightarrow 0 \quad \text{in probability,}$$

noting $\Lambda[F_n]/(m-1)! = \mathcal{Y}(\xi_n)$ by Definitions 5.1 and 6.2. Finally $\xi_n \rightarrow \xi$ in law in $C(M)$ (Theorem 2.4) and $\mathcal{Y}$ is continuous (Lemma 6.3), so $\mathcal{Y}(\xi_n) \rightarrow \mathcal{Y}(\xi)$ in law by the continuous mapping theorem [1, Thm. 5.1]; conclude by Slutsky's theorem [1, Thm. 5.2(2)]. $\square$

Corollary 6.5 (Free energy formula; cf. [1, Main Thm. 6.2(1), Cor. 6.1(1)]). *In law,*

$$F_n^0 - \lambda \log n + (m-1) \log \log n \longrightarrow -\log \mathcal{Y}(\xi),$$

and consequently

$$F_n = n\beta S_n + \lambda \log n - (m-1) \log \log n + F_n^R, \quad F_n^R \longrightarrow -\log \mathcal{Y}(\xi) \text{ in law.}$$

Proof. $\mathcal{Y}(\xi) \in (0, \infty)$ almost surely (Lemma 6.3 applied to the continuous paths of ξ), and $-\log$ is continuous on $(0, \infty)$; apply the continuous mapping theorem to Theorem 6.4, and use $F_n = n\beta S_n + F_n^0$. $\square$

No essential family of coordinates, no chart constant γ_b , and no partition of unity appears in Theorem 6.4 or Corollary 6.5: the limit is a functional of the pair $(\mu, \xi|_\Sigma)$ together with the invariants (λ, m) . (The pair depends on the chosen resolution; the law of $\mathcal{Y}(\xi)$ does not — see Remark 8.1.)

7 Convergence of expectations

In this section Assumption 2.1(I) is strengthened to index $s = 4$; all other notation and standing data are as before. The goal is the expectation version of Corollary 6.5, i.e. the coordinate-free form of [1, Main Thm. 6.2(2), Cor. 6.1(2)]. Throughout, write

$$A_n := F_n^0 - \lambda \log n + (m-1) \log \log n = -\log \left(\frac{n^\lambda}{(\log n)^{m-1}} Z_n^0 \right) \quad (n \geq 3),$$

so that Corollary 6.5 states $A_n \rightarrow -\log \mathcal{Y}(\xi)$ in law.

7.1 Background: asymptotic uniform integrability

Definition 7.1 ([1, Def. 5.2]). A sequence of real random variables $\{X_n\}$ is *asymptotically uniformly integrable* (AUI) if

$$\lim_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \sup_{N \geq n} \mathbb{E}[|X_N| \mathbf{1}\{|X_N| \geq M\}] = 0.$$

Theorem 7.2 ([1, Thms. 5.3–5.5]). (a) *If $|X_n| \leq Y_n$ with $\{Y_n\}$ AUI, then $\{X_n\}$ is AUI; the property is preserved under multiplication by a constant (immediate from Definition 7.1).*

(b) *If $\sup_n \mathbb{E}[X_n^2] < \infty$, then $\{X_n\}$ is AUI ([1, Thm. 5.4(3)] with $s = 2$, $\delta = 1$).*

(c) *If $X_n \rightarrow X$ in law, $\sup_n \mathbb{E}[|X_n|] < \infty$, and $\{X_n\}$ is AUI, then $\mathbb{E}[|X|] < \infty$ and $\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$.*

7.2 Fourth moments

Lemma 7.3. *Under Assumption 2.1 with index $s = 4$,*

$$\sup_n \mathbb{E} \left[\sup_M |\xi_n|^4 \right] < \infty, \quad \sup_n \mathbb{E} \left[\sup_{\{K \geq \epsilon\}} |\psi_n|^4 \right] < \infty.$$

Proof. Under (I) at index $s = 4$, the function $a(\cdot, u)$ of Theorem 2.3 is $L^4(q)$ -valued analytic on each chart ([1, Thm. 6.1] applied at index 4), and the family $w \mapsto (K(w) - f(\cdot, w))/\sqrt{K(w)}$ defining ψ_n is $L^4(q)$ -valued analytic on a neighborhood of the compact set $\{K \geq \epsilon\}$, since $K \geq \epsilon > 0$ there (cf. the proof of [1, Thm. 6.2]). Apply [1, Thm. 5.8] with $s = 4$ on the finite atlas, resp. on a finite cover of $\{K \geq \epsilon\}$. Only these uniform bounds are needed below; the convergence of the corresponding moments ([1, Thm. 6.2(2)]) is not used. $\square$

7.3 A two-sided envelope for A_n

Set $V_n := \max(\|\xi_n\|_\infty^2, \sup_{\{K \geq \epsilon\}} \psi_n^2)$.

Lemma 7.4 (Envelope). *There exist a deterministic n_0 and a constant C_0 such that, surely, for all $n \geq n_0$:*

$$(i) \quad A_n \leq \frac{\beta}{2} \|\xi_n\|_\infty^2 + C_0;$$

$$(ii) \quad A_n \geq -\frac{\beta}{2} V_n - C_0.$$

In particular $|A_n| \leq \frac{\beta}{2} V_n + C_0$ for $n \geq n_0$.

Proof. (i) (Lower bound on the evidence.) Pointwise on M , with $t = nK_g(u)$, Cauchy–Schwarz gives $\beta\sqrt{t}\xi_n(u) \geq -\frac{\beta}{2}t - \frac{\beta}{2}\|\xi_n\|_\infty^2$, so

$$Z_n^0 \geq Z_n^{(1)} \geq e^{-\beta\|\xi_n\|_\infty^2/2} \int_M e^{-\frac{3}{2}n\beta K_g(u)} \nu(du).$$

The last integral equals $\frac{(\log n)^{m-1}}{(m-1)!n^\lambda} \Lambda_n[F]$ for the deterministic insertion $F(t, u) = e^{-3\beta t/2}$, which is admissible with $A = 0$ and $N_0(F) = 1$; by Theorem 5.2 and Proposition 3.2,

$$\Lambda_n[F] \longrightarrow \Lambda[F] = \Gamma(\lambda) \left(\frac{2}{3\beta}\right)^\lambda \mu(M) > 0.$$

Choose n_0 so that $\Lambda_n[F] \geq c_0$ for $n \geq n_0$, with $c_0 := \frac{1}{2}\Lambda[F] > 0$; then $A_n \leq \frac{\beta}{2}\|\xi_n\|_\infty^2 + \log((m-1)!/c_0)$.

(ii) (Upper bound on the evidence.) Pointwise, $\beta\sqrt{t}\xi_n \leq \frac{\beta}{2}t + \frac{\beta}{2}\|\xi_n\|_\infty^2$, so by Lemma 4.1 with the admissible $G(t) = e^{-\beta t/2}$,

$$Z_n^{(1)} \leq e^{\beta\|\xi_n\|_\infty^2/2} \int_M e^{-\frac{1}{2}n\beta K_g} d\nu \leq C e^{\beta\|\xi_n\|_\infty^2/2} \frac{(\log n)^{m-1}}{n^\lambda},$$

while on the outer region, as in the proof of Theorem 6.4, $Z_n^{(2)} \leq e^{-n\beta\epsilon/2} e^{\frac{\beta}{2}\sup_{\{K \geq \epsilon\}} \psi_n^2}$. Enlarging n_0 so that $n^\lambda(\log n)^{1-m}e^{-n\beta\epsilon/2} \leq 1$, and using $Ce^p + e^q \leq (C+1)e^{\max(p,q)}$,

$$\frac{n^\lambda}{(\log n)^{m-1}} Z_n^0 \leq C' \exp\left(\frac{\beta}{2} V_n\right),$$

and (ii) follows by taking $-\log$. $\square$

Remark 7.5. Part (ii) is the coordinate-free counterpart of the estimate in the proof of [1, Main Thm. 6.2(2)]. Part (i) replaces the chart-level lower bounds [1, Thms. 4.8, 4.10]: in the present framework the lower bound on the evidence is not a separate computation but an instance of the kernel theorem for a deterministic insertion, combined with the positivity $\mu(M) > 0$. The upper-bound-only character of §§4–5 is thereby sufficient after all.

7.4 The expectation version of the free energy formula

Theorem 7.6 (Expectation version; cf. [1, Main Thm. 6.2(2), Cor. 6.1(2)]). *Assume (I) with index $s = 4$ and (II). Then $\{A_n\}_{n \geq n_0}$ is AUI with $\sup_{n \geq n_0} \mathbb{E}[|A_n|] < \infty$, the expectation $\mathbb{E}[|\log \mathcal{Y}(\xi)|]$ is finite, and*

$$\mathbb{E}[F_n^0] - \lambda \log n + (m-1) \log \log n \longrightarrow -\mathbb{E}[\log \mathcal{Y}(\xi)].$$

Consequently, if the entropy $S = -\int q \log q \, dx$ is finite,

$$\mathbb{E}[F_n] = n\beta S + \lambda \log n - (m-1) \log \log n - \mathbb{E}[\log \mathcal{Y}(\xi)] + o(1).$$

Proof. By Lemma 7.4, $|A_n| \leq \frac{\beta}{2} V_n + C_0 \leq C_1(1 + V_n)$ for $n \geq n_0$. By Lemma 7.3 and $V_n^2 \leq \|\xi_n\|_\infty^4 + \sup_{\{K \geq \epsilon\}} \psi_n^4$, we have $\sup_n \mathbb{E}[(1 + V_n)^2] < \infty$; hence $\{1 + V_n\}$ is AUI by Theorem 7.2(b), and $\{A_n\}_{n \geq n_0}$ is AUI by Theorem 7.2(a), with $\mathbb{E}[|A_n|] \leq C_1(1 + \mathbb{E}[V_n])$ uniformly bounded. By Corollary 6.5, $A_n \rightarrow -\log \mathcal{Y}(\xi)$ in law; Theorem 7.2(c) then yields $\mathbb{E}[|\log \mathcal{Y}(\xi)|] < \infty$ and $\mathbb{E}[A_n] \rightarrow -\mathbb{E}[\log \mathcal{Y}(\xi)]$, which is the first display. The second follows from $F_n = n\beta S_n + F_n^0$ and $\mathbb{E}[S_n] = S$. $\square$

Remark 7.7. The integrability of $\log \mathcal{Y}(\xi)$ can also be read off directly and two-sidedly: the Cauchy–Schwarz estimates used above give, for every $\eta \in C(M)$,

$$\frac{\mu(M)}{(m-1)!} e^{-\beta \|\eta\|_\infty^2/2} \int_0^\infty t^{\lambda-1} e^{-3\beta t/2} dt \leq \mathcal{Y}(\eta) \leq \frac{\mu(M)}{(m-1)!} e^{\beta \|\eta\|_\infty^2/2} \int_0^\infty t^{\lambda-1} e^{-\beta t/2} dt,$$

so $|\log \mathcal{Y}(\eta)| \leq \frac{\beta}{2} \|\eta\|_\infty^2 + C_Y$; and $\|\xi\|_\infty^2$ has moments of every order by Fernique’s theorem [7].

8 Remarks

Remark 8.1 (Invariance, and the real log canonical threshold). The pair (λ, m) is intrinsic to $(K, \varphi \, dw)$ on W : it is read off the zeta function downstairs (§2), before any resolution is chosen. When $\varphi > 0$ on a neighborhood of W_0 , λ is the *real log canonical threshold* of K (along W_0) and m its multiplicity; in general (λ, m) is the real log canonical threshold data of the pair $(K, \varphi \, dw)$, the zeros of φ_2 entering through the multi-indices h .

By contrast, μ , Σ , and the process ξ live on M and depend on the chosen resolution. The pushforward $g_*\mu$ is a canonical positive measure on W_0 : by (3), for $\Psi \in C(W_\epsilon)$,

$$(g_*\mu)[\Psi] = \mu[\Psi \circ g] = \lim_{\sigma \rightarrow -\lambda^+} (\sigma + \lambda)^m \int_{W_\epsilon} K(w)^\sigma \Psi(w) \varphi(w) \, dw,$$

a limit in which no resolution appears. But the pair $(g_*\mu, \cdot)$ does not suffice downstairs: the function $a(x, \cdot)$, hence ξ , genuinely varies along positive-dimensional fibres of g , so the limit functional cannot in general be expressed on W . What is resolution-independent is the *law* of $\mathcal{Y}(\xi)$, since by Theorem 6.4 it is the limit in law of the resolution-free quantities $n^\lambda (\log n)^{1-m} Z_n^0$.

Remark 8.2 (What has been replaced). Definition 3.1 and Proposition 3.2 replace [1, Thm. 4.6] together with the consistency bookkeeping of [1, Thm. 6.5(3)–(4)]; Theorem 5.2 replaces [1, Thms. 4.7–4.9] and the chart-by-chart assembly in the proof of [1, Thm. 6.7], and covers all insertions at once. Convexity of the orthant charts makes the interpolation-to-the-face step

elementary and avoids constructing a global C^1 retraction onto Σ (which would face gluing questions where essential faces meet). Charts persist inside the proofs of Proposition 3.2, Lemmas 4.1–4.2 and Theorem 5.2, where the exact monomial evaluations of [1, Thm. 4.6, Cor. 4.3] are cited; this appears irreducible, the evaluation being a monomial computation. Two smaller repairs are worth recording. First, the error bound of [1, Thm. 4.9] involves $\|\nabla\xi\|$, but tightness of $\|\nabla\xi_n\|$ is nowhere established in [1]; Lemma 6.1 supplies it. Second, under the dictionary of Remark 3.3, $\mathcal{Y}(\xi)$ coincides with the chartwise limit $Y^{(1)}(\xi)$ appearing in the proof of [1, Thm. 6.7]; the limit displayed in the *statement* of [1, Thm. 6.7] omits the monomial factor y^μ that its own proof (and [1, Thm. 4.9]) carries.

Remark 8.3 (Next steps). With the expectation statements of Section 7 in place, the remaining program is the four error formulas (Bayes/Gibbs, generalization/training), which are expectations of functionals of the renormalized posterior kernel

$$t^{\lambda-1} e^{-\beta t + \beta \sqrt{t} \xi(u)} d\mu(u) dt \quad \text{on } (0, \infty) \times \Sigma,$$

with insertions of the form t^p and $\sqrt{t}a(x, u)$ — all within the scope of Theorem 5.2, which is why it was stated for general $F(t, u)$. The singular fluctuation of [1, Def. 5.9] is already formulated in exactly this language: a measure $\mu(du)$ on a compact subset of a manifold, a Gaussian process with $\mathbb{E}[\xi(u)^2] = 2$ there (Theorem 2.4).

Remark 8.4 (Literature). The scheme “coefficients of the asymptotic expansion of $\int e^{-nf} \varphi$ are distributions supported on the critical locus, realized as Laurent coefficients of $\int f^z \varphi$, with orders controlled by the drop in pole order” is developed in Arnold–Gusein-Zade–Varchenko [3, Vol. II, Part II], including Varchenko’s computation of (λ, m) from Newton polyhedra in nondegenerate cases [4]; two caveats when consulting these sources are that the amplitudes there are usually nonvanishing (the monomial density $u^h \phi$ shifts the numerology exactly as in (8)), and that much of the theory is stated for isolated critical points. Malgrange [5] connects the pole structure to the Bernstein–Sato polynomial, i.e. to [1, §4.5]; Barlet [6] is the complex-analytic counterpart, where the positivity of leading coefficients appears in the theory of the currents associated with $\int |f|^{2z}$.

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